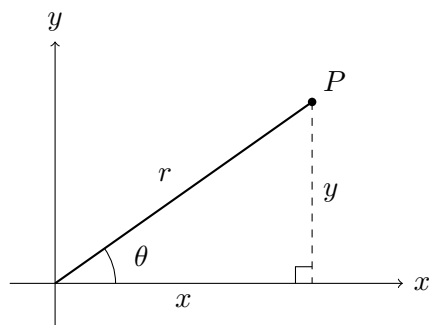


Converting Between Rectangular and Polar Coordinates

Polar to rectangular, rectangular to polar, choosing the angle by quadrant, and naming one point in more than one way

The picture behind the formulas. Drop a perpendicular from the point P to the horizontal axis. The right triangle it makes has legs x and y and hypotenuse r , with the angle θ at the origin. Every conversion formula below is read straight off this triangle. *No values are shown here — use the numbers given in each problem.*



Directions.

- Polar $\rightarrow$ rectangular: $x = r \cos \theta$, $y = r \sin \theta$. These always work, including when r is negative.
- Rectangular $\rightarrow$ polar: $r = \sqrt{x^2 + y^2}$, and θ from $\tan \theta = \frac{y}{x}$ — but the calculator's $\tan^{-1}$ only reports angles between -90° and 90° , so **check the quadrant and add 180° when $x < 0$.**
- Give exact values (radicals, fractions) unless a problem asks for decimals. Use $0^\circ \leq \theta < 360^\circ$ unless told otherwise.

1. A point has polar coordinates $(6, 60^\circ)$. Find its rectangular coordinates exactly.

(a) $x =$

Answer: _____

(b) $y =$

Answer: _____

2. A point has rectangular coordinates $(3, 3)$. Find polar coordinates with $r > 0$ and $0^\circ \leq \theta < 360^\circ$.

(a) $r =$

Answer: _____

(b) $\theta =$

Answer: _____ degrees

3. A point has polar coordinates $(8, 210^\circ)$. Find its rectangular coordinates exactly.

(a) $x =$

Answer: _____

(b) $y =$

Answer: _____

4. A point has rectangular coordinates $(5, 12)$.

(a) Find r exactly.

Answer: _____

(b) Find θ to two decimal places.

Answer: _____ degrees

5. A point has rectangular coordinates $(-2, 2\sqrt{3})$.

(a) Find r exactly.

Answer: _____

(b) Find θ with $0^\circ \leq \theta < 360^\circ$.

Answer: _____ degrees

(c) A calculator returns -60° for $\tan^{-1}\left(\frac{2\sqrt{3}}{-2}\right)$. Explain why that is not the answer, and why your correction is the right one.

Answer: _____

6. A point has polar coordinates $(-4, 45^\circ)$ — note the negative r . Find its rectangular coordinates exactly.

(a) $x =$

Answer: _____

(b) $y =$

Answer: _____

7. A point has polar coordinates $\left(5, \frac{2\pi}{3}\right)$, with the angle in radians. The same point can be named with a negative first coordinate, as $(-5, \theta')$.

(a) Find θ' with $0 \leq \theta' < 2\pi$.

Answer: _____

(b) Compute $x = -5 \cos \theta'$ to confirm that both names give the same point.

Answer: _____

8. A point has rectangular coordinates $(-6, -6)$.

(a) Find r exactly.

Answer: _____

(b) Find θ with $0^\circ \leq \theta < 360^\circ$.

Answer: _____ degrees

9. A point has rectangular coordinates $(0, -7)$.

(a) Find r .

Answer: _____

(b) The angle is $\theta = 270^\circ$. Compute $7 \sin 270^\circ$ to check that this angle really returns the given y .

Answer: _____

(c) Explain why the $\tan^{-1}$ formula cannot be used for this point at all, and how the angle is found instead.

Answer: _____

10. On a radar screen the station sits at the origin and distances are in kilometres. An aircraft is at rectangular coordinates $(-9, 12)$.

(a) Find its distance r from the station.

Answer: _____ km

(b) Find θ with $0^\circ \leq \theta < 360^\circ$, to two decimal places.

Answer: _____ degrees

(c) The same aircraft can be logged with a negative first coordinate, as $(-15, \theta'')$. Find θ'' with $0^\circ \leq \theta'' < 360^\circ$, to two decimal places.

Answer: _____ degrees

(d) Explain why $(-15, \theta'')$ describes the same aircraft as $(15, \theta)$.

Answer: _____